

Brazilian Economic Pipeline

Airflow · PySpark · Delta Lake · dbt · DuckDB · Streamlit

DELIVERY STATUS

- ✓ Complete and functional pipeline
- ✓ All dbt tests passing
- ✓ Streamlit dashboard operational
- ✓ Full documentation included

What Was Delivered

Ingestion Pipeline ingestion/apis.py

Ingests real data from IBGE, Central Bank and IPEA with automatic fallback. If IBGE returns a 500 error, GDP data is automatically fetched from the Central Bank.

Delta Lake Bronze/Silver spark_jobs/ (bronze, silver, bridge)

PySpark writes raw data to Bronze and transforms to Silver with ACID MERGE. Automatic deduplication — reprocessing does not accumulate duplicates.

Gold Layer (dbt) dbt_project/models/gold/

3 SQL models with documented lineage: annual macro panel, monthly series with YoY and correlations.

Airflow Orchestration airflow/dags/econ_pipeline.py

DAG with 3 parallel TaskGroups (IBGE, BCB, IPEA), ShortCircuitOperator for graceful failure and fan-in on Silver.

Dashboard dashboard/app.py

4 analytical pages: Macro Panel, Time Series, Correlations and Pipeline Architecture.

Docker Infrastructure docker/Dockerfile + docker-compose.yml

Full containerized stack: Airflow + Postgres + Java + PySpark + dbt in a single docker compose up.

How to Run

1. Install dependencies

```
pip install -r requirements.txt
```

2. Install dbt packages (required, only once)

```
cd dbt_project
dbt deps
cd ..
```

dbt deps downloads the dbt-utils package declared in packages.yml. Without this step, dbt run fails. Only needs to run once — or after deleting the dbt_packages/ folder.

3. Run the full pipeline

```
python run.py
```

This runs ingestion → Bronze → Silver → Bridge → dbt run → dbt test → dashboard. Streamlit opens automatically at <http://localhost:8501>.

Additional options

<code>python run.py --only-pipeline</code>	Pipeline only, without opening the dashboard
<code>python run.py --only-dashboard</code>	Dashboard only (pipeline already ran)
<code>python run.py --skip-spark</code>	Skip Spark — reprocess only dbt
<code>python run.py --fontes bcb_selic</code>	Process only specific sources

Pipeline Architecture

```
APIs → Bronze (Delta Lake) → Silver (ACID MERGE) → DuckDB (bridge) → dbt Gold → Streamlit
```

Layer	Technology	Detail
Bronze	PySpark + Delta Lake	Raw data, partitioned by source/year
Silver	PySpark + Delta Lake	Cleaned, deduplicated, ACID MERGE on every run
Bridge	PySpark + DuckDB	Exports Parquet → registers view in DuckDB
Gold	dbt + DuckDB	3 SQL models with automatic quality tests
Dashboard	Streamlit + Plotly	Reads gold layer via DuckDB read-only

Common Troubleshooting

Issue: dbt run fails with `KeyError snapshot_helper.sql`

Fix: `cd dbt_project && rm -rf dbt_packages target package-lock.yml && dbt deps`

Issue: Invalid packages.yml

Fix: File must contain only: `- package: dbt-labs/dbt_utils / version: 1.3.0`

Issue: `SerializationException` in DuckDB

Fix: Delete `data/warehouse/econ.duckdb` and run: `python run.py --skip-spark`

Issue: None columns in dashboard (GDP, income)

Fix: Delete `data/delta/silver` and `data/warehouse/econ.duckdb`, run: `python run.py --only-pipeline`